Three Decades of Pediatric NHANES Research: Methodological Trends, Thematic Evolution, and Missed Opportunities in Biomarker Utilization: A Bibliometric Analysis

1. PRISMA NUMBERS

Stage	N
Records identified (Web of Science)	4574
After document type restriction (Article/Review)	4458
After duplicate removal (DOI, then Title)	4457
After year correction (PY ≤ 2025) and early access removal	4445
Final records included (after removing records without abstract)	4435

2. TOP FIFTEEN AUTHORS

Author	Publications
FULGONI VL	77
OGDEN CL	46
NICKLAS TA	42
JAIN RB	40
KIM SH	40
PARK S	37
KIM JH	36
HAN K	33
O'NEIL CE	33
BAILEY RL	31
REHM CD	30
AUINGER P	29
KIM DH	29
ZHANG J	29
DREWNOWSKI A	28

3. TOP 20 JOURNALS BY PUBLICATION COUNT

Journal	Publications
NUTRIENTS	156
PEDIATRICS	111
PLOS ONE	104
AMERICAN JOURNAL OF CLINICAL NUTRITION	103
ENVIRONMENTAL RESEARCH	79
JOURNAL OF NUTRITION	78
ENVIRONMENTAL HEALTH PERSPECTIVES	77
JOURNAL OF PEDIATRICS	61
PUBLIC HEALTH NUTRITION	60
SCIENTIFIC REPORTS	59
INTERNATIONAL JOURNAL OF ENVIRONMENTAL RESEARCH AND PUBLIC HEALTH	54
JOURNAL OF THE ACADEMY OF NUTRITION AND DIETETICS	53
FRONTIERS IN NUTRITION	50
ENVIRONMENTAL SCIENCE AND POLLUTION RESEARCH	47
AMERICAN JOURNAL OF PREVENTIVE MEDICINE	45
BMC PUBLIC HEALTH	43
PEDIATRIC OBESITY	40
JAMA-JOURNAL OF THE AMERICAN MEDICAL ASSOCIATION	39
AMERICAN JOURNAL OF HUMAN BIOLOGY	38
SCIENCE OF THE TOTAL ENVIRONMENT	35

4. **Top 15 institutions by publication count**

Institution	Publications
CENTERS FOR DISEASE CONTROL AND PREVENTION - USA	605
HARVARD UNIVERSITY	446
UNIVERSITY OF CALIFORNIA SYSTEM	322
NATIONAL INSTITUTES OF HEALTH (NIH) - USA	296
JOHNS HOPKINS UNIVERSITY	285
SEOUL NATIONAL UNIVERSITY (SNU)	276
UNIVERSITY OF NORTH CAROLINA	261
YONSEI UNIVERSITY	234
UNIVERSITY OF NORTH CAROLINA CHAPEL HILL	222
UNIVERSITY SYSTEM OF OHIO	210
PENNSYLVANIA COMMONWEALTH SYSTEM OF HIGHER EDUCATION (PCSHE)	192
CDC NATIONAL CENTER FOR HEALTH STATISTICS (NCHS)	187
HARVARD T.H. CHAN SCHOOL OF PUBLIC HEALTH	184
HARVARD UNIVERSITY MEDICAL AFFILIATES	175
NEW YORK UNIVERSITY	170

5. **Top 15 countries by author-country affiliations (each paper counted once per author per country).**

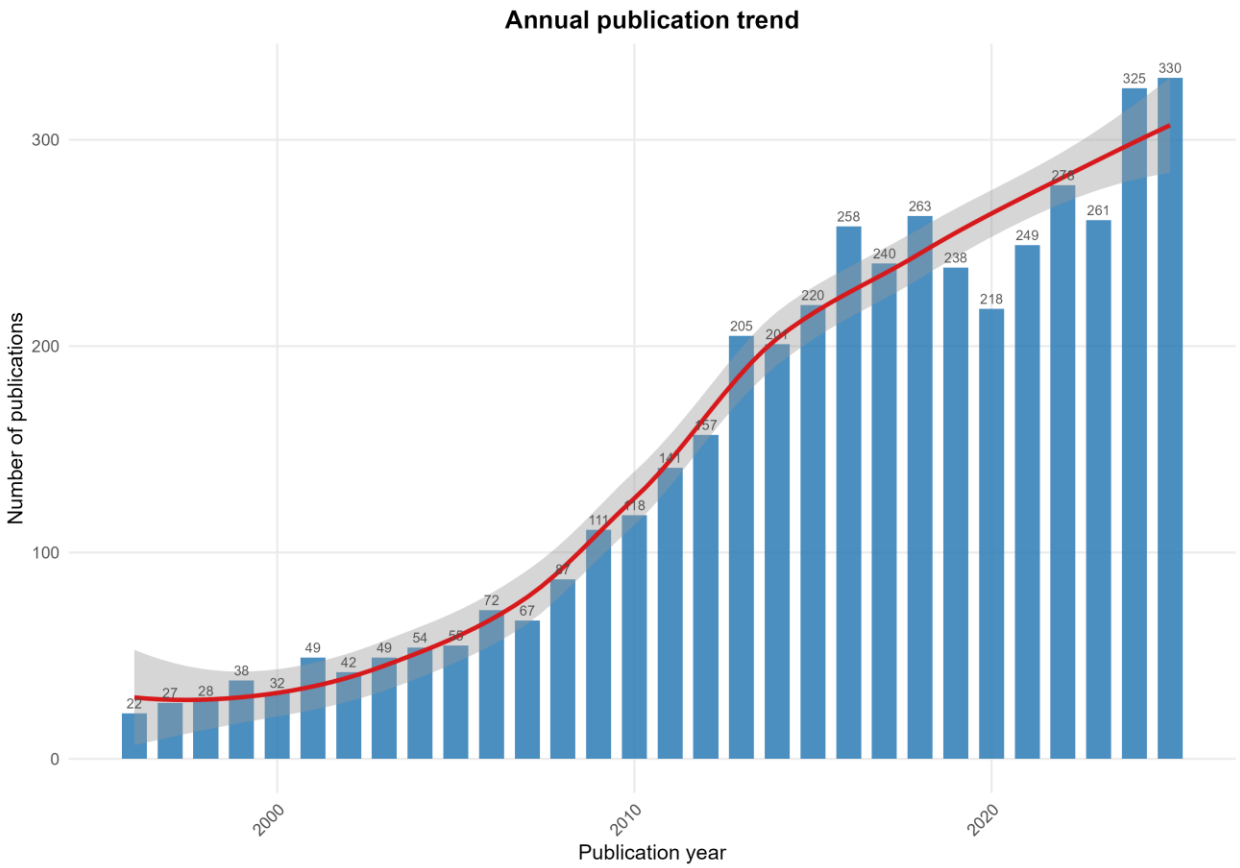
Country	Author.country.affiliations
USA	9019
CHINA	1908
KOREA	1707
CANADA	352
UNITED KINGDOM	174
AUSTRALIA	136
GERMANY	105
BRAZIL	95
SPAIN	83
ARGENTINA	65
IRAN	49
ITALY	48
SOUTH AFRICA	47
FRANCE	45
MEXICO	44

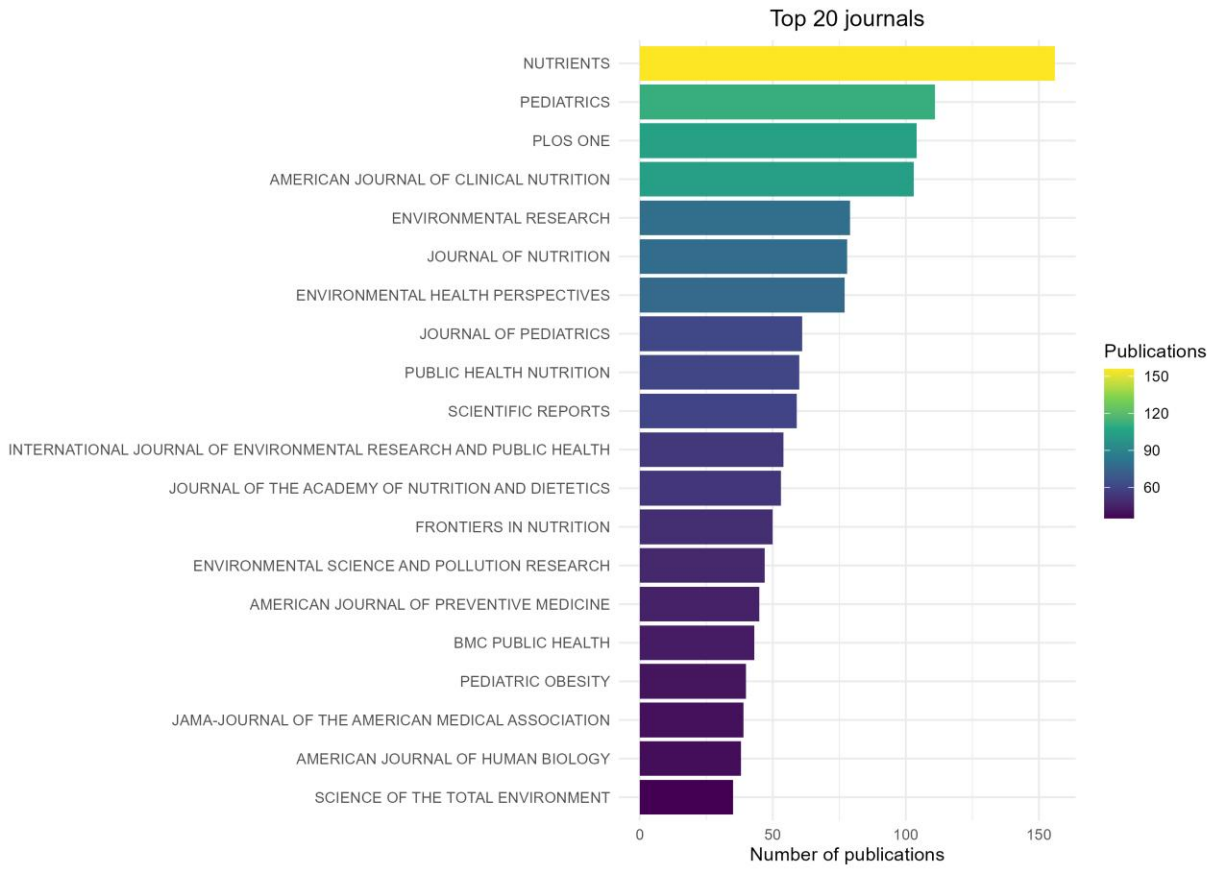
Biomarker Freq

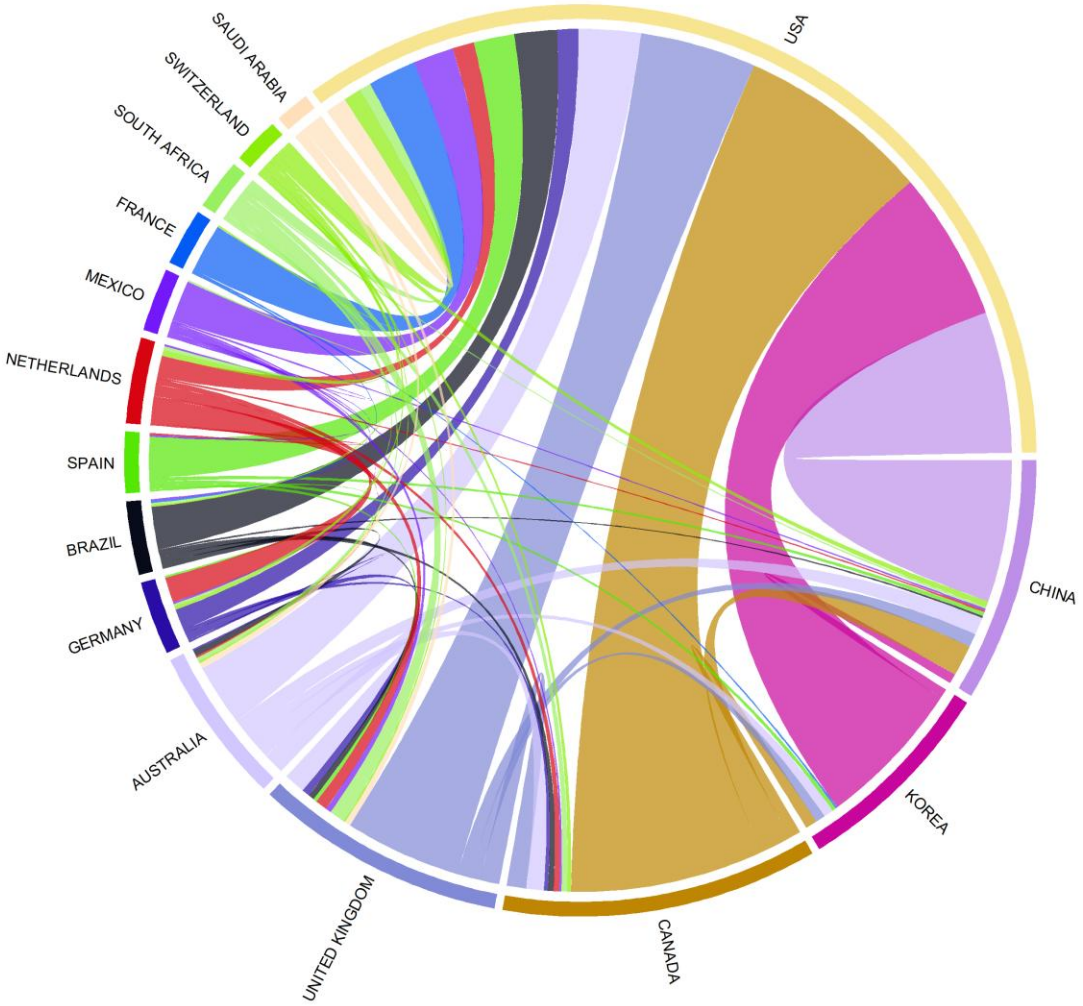
Biomarker	Count
Lead / blood lead	471
Cholesterol	458
Insulin	353
Glucose	281
Triglycerides	269
Vitamin D / 25(OH)D	239
Cotinine	174
Creatinine	160
Mercury	159
Cadmium	141
CRP / C-reactive protein	140
PFAS / perfluoro	100
HbA1c	68
eGFR	50

Biomarkers with fewer than 100 mentions

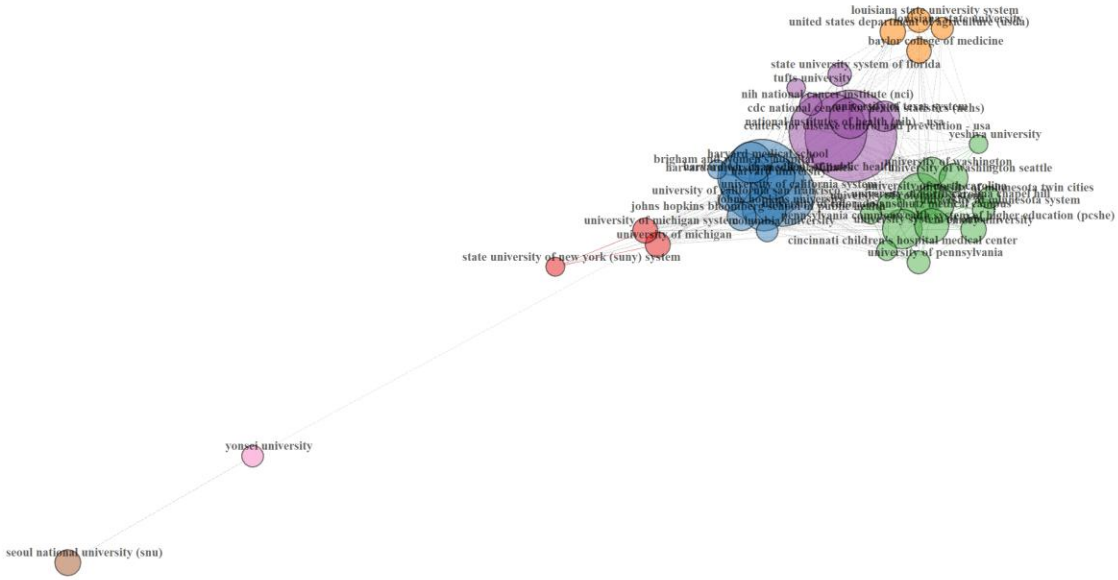
Biomarker	**Count**
HbA1c	68
eGFR	50


Annual publication trend
Number of publications
Publication year
0
100
200
300
2000
2010
2020
22
27
28
38
32
49
42
49
54
58
72
67
87
111
118
141
157
205
201
220
258
240
263
238
218
249
276
261
325
330

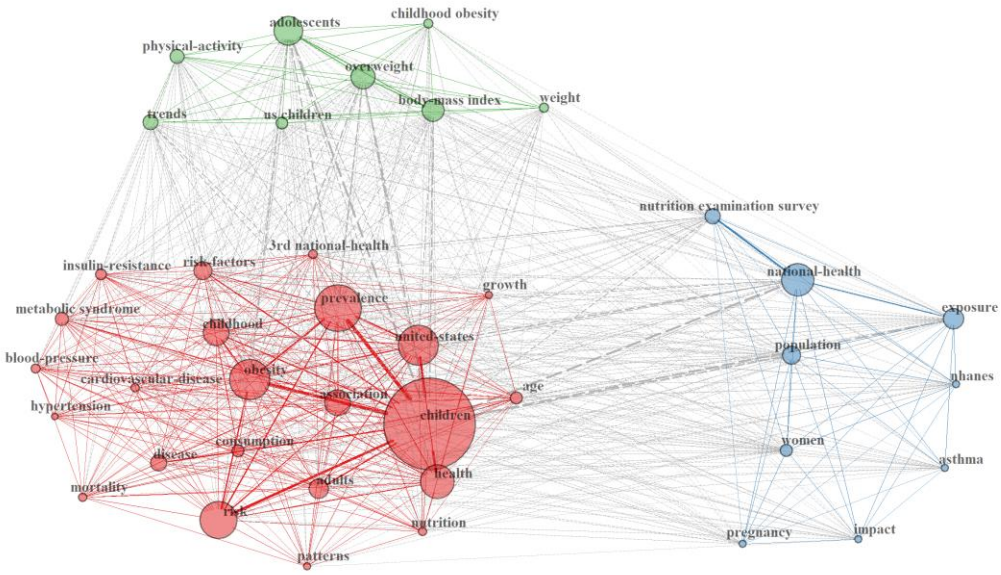

Top 20 journals
NUTRIENTS
PEDIATRICS
PLOS ONE
AMERICAN JOURNAL OF CLINICAL NUTRITION
ENVIRONMENTAL RESEARCH
JOURNAL OF NUTRITION
ENVIRONMENTAL HEALTH PERSPECTIVES
JOURNAL OF PEDIATRICS
PUBLIC HEALTH NUTRITION
SCIENTIFIC REPORTS
INTERNATIONAL JOURNAL OF ENVIRONMENTAL RESEARCH AND PUBLIC HEALTH
JOURNAL OF THE ACADEMY OF NUTRITION AND DIETETICS
FRONTIERS IN NUTRITION
ENVIRONMENTAL SCIENCE AND POLLUTION RESEARCH
AMERICAN JOURNAL OF PREVENTIVE MEDICINE
BMC PUBLIC HEALTH
PEDIATRIC OBESITY
JAMA-JOURNAL OF THE AMERICAN MEDICAL ASSOCIATION
AMERICAN JOURNAL OF HUMAN BIOLOGY
SCIENCE OF THE TOTAL ENVIRONMENT
0
50
100
150
Number of publications
Publications
150
120
90
60

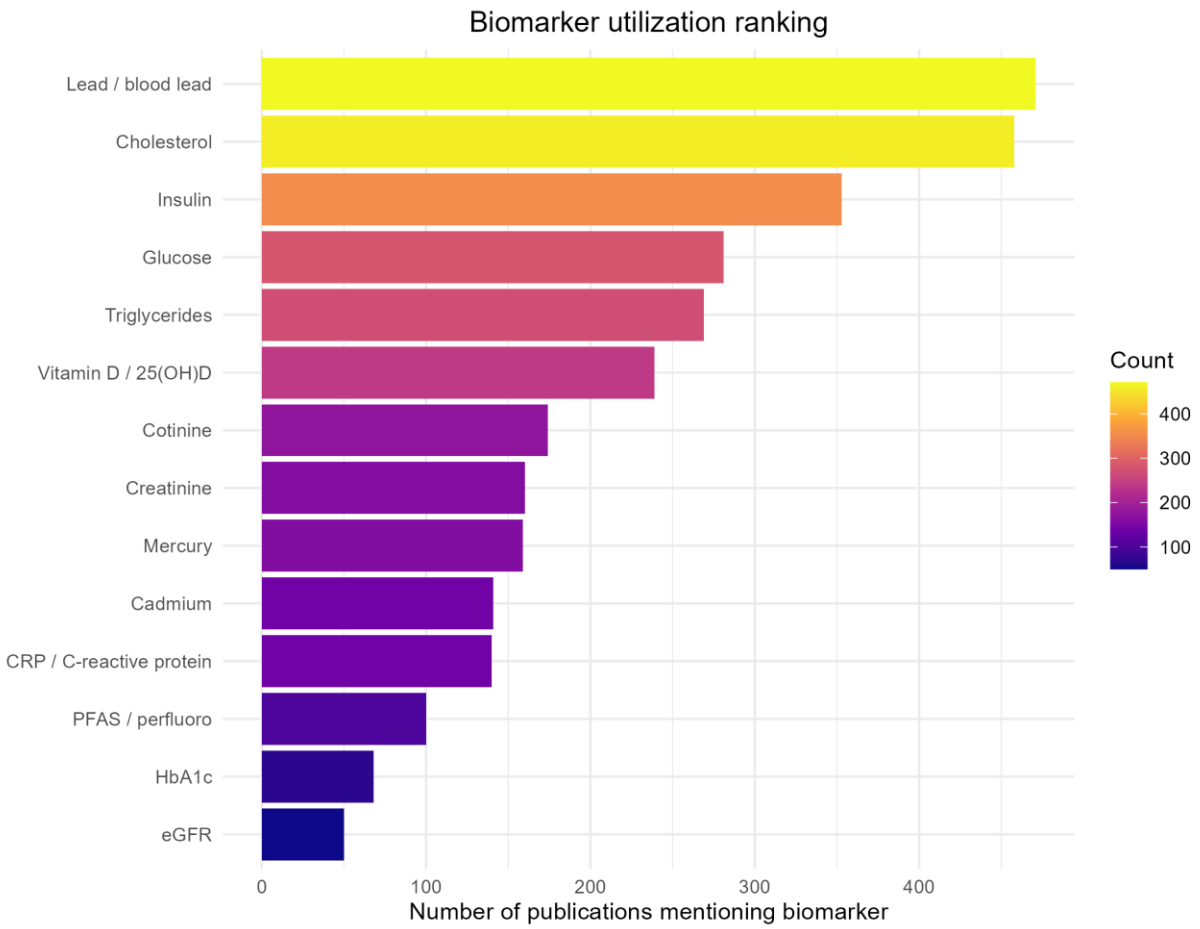

USA
CHINA
KOREA
CANADA
UNITED KINGDOM
AUSTRALIA
GERMANY
BRAZIL
SPAIN
NETHERLANDS
MEXICO
FRANCE
SOUTH AFRICA
SWITZERLAND
SAUDI ARABIA

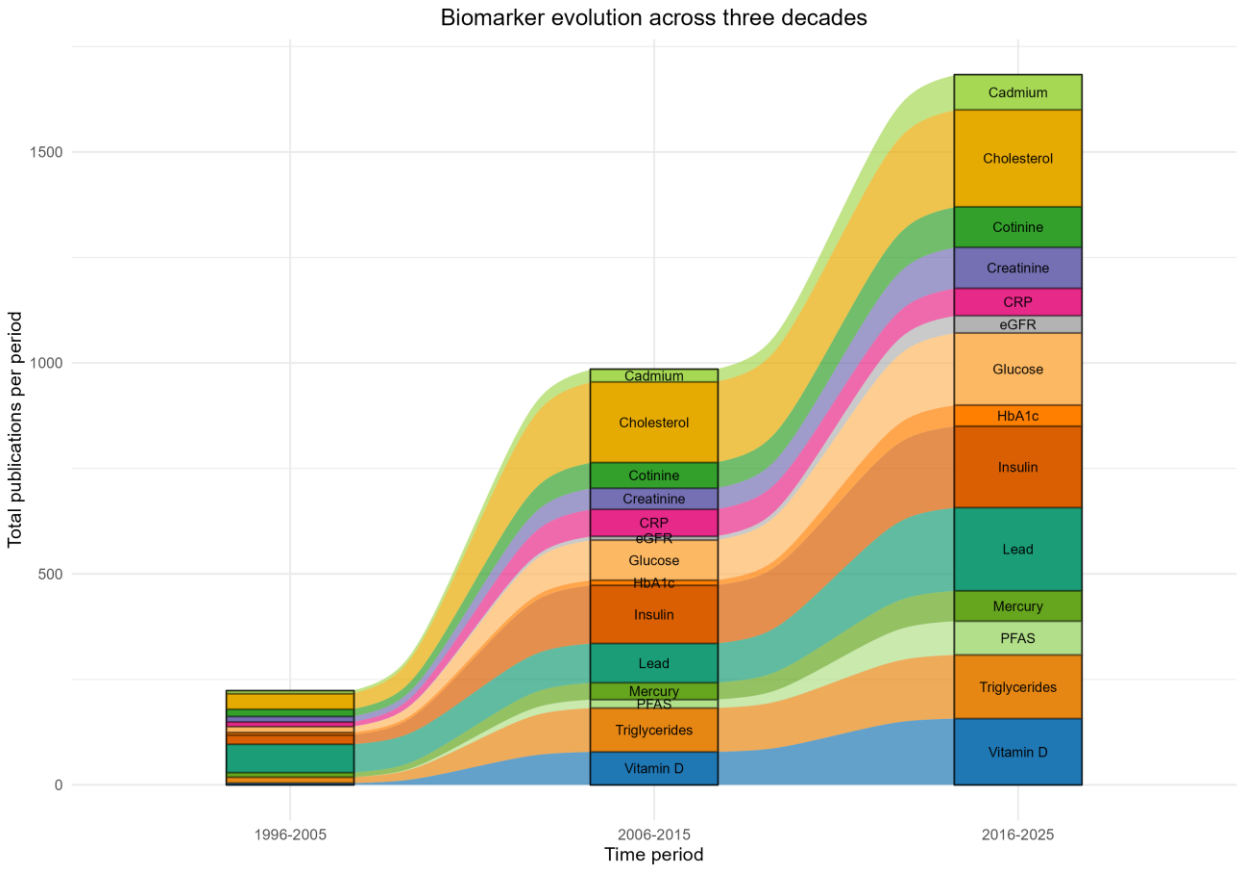
Plot

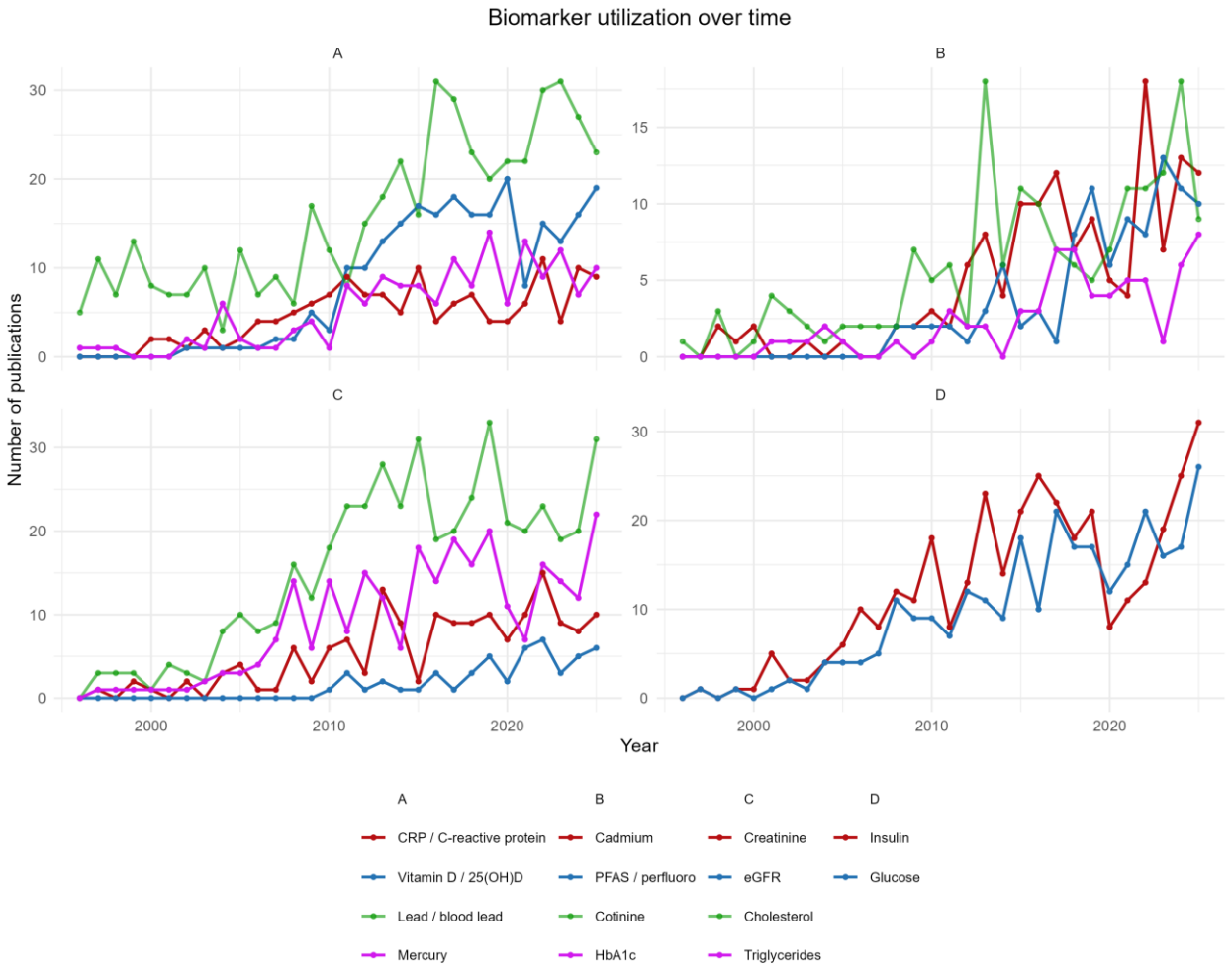

louisiana state university system
baylor college of medicine
state university system of florida
tufts university
nih national cancer institute (nci)
yeshiva university
university of washington seattle
university of california system
johns hopkins bloomberg school of public health
university of michigan system
university of michigan
cincinnati children's hospital medical center
university of pennsylvania
state university of new york (suny) system
yonsei university
seoul national university (snu)

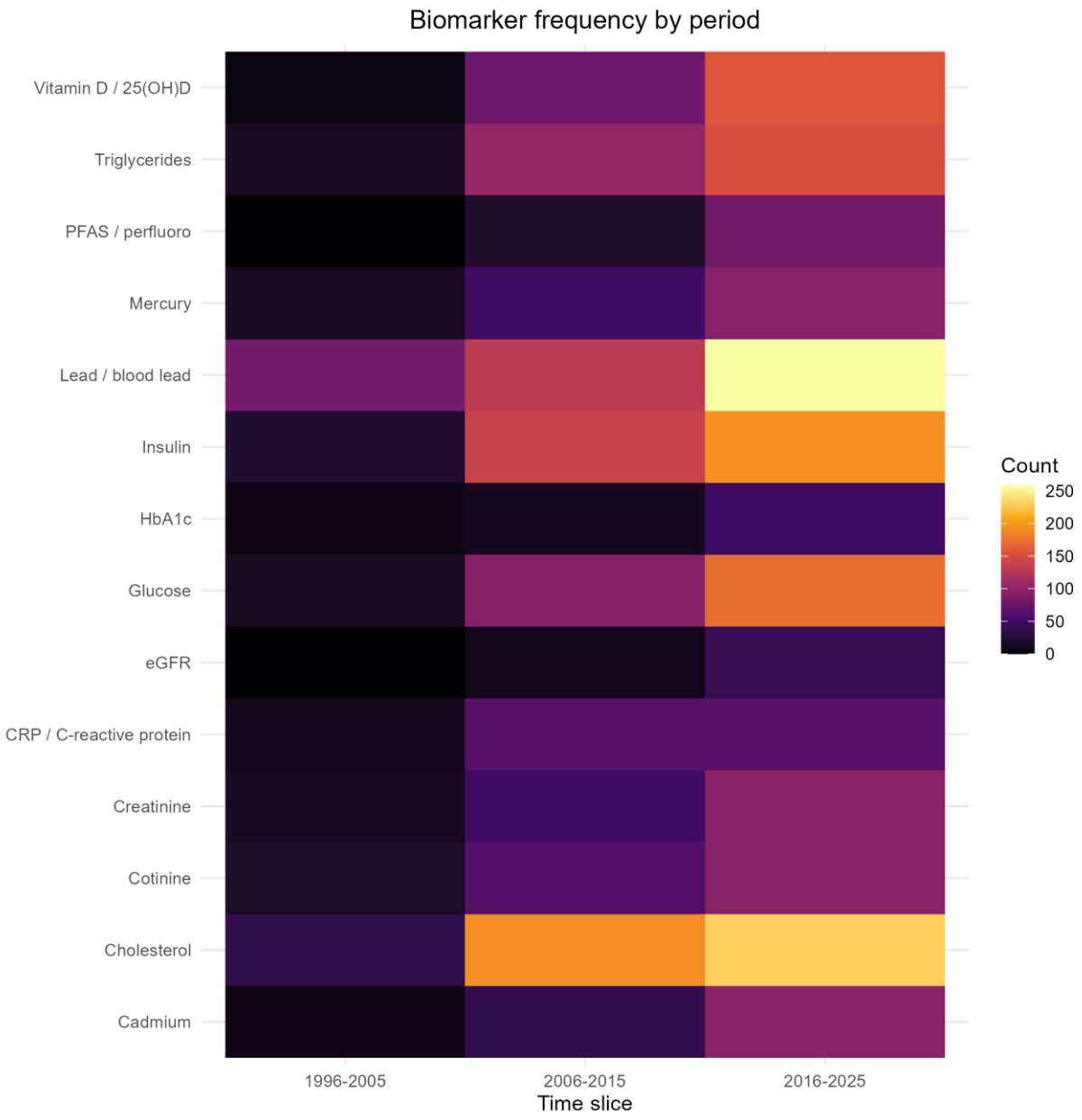
Plot


adolescents
childhood obesity
physical-activity
overweight
body-mass index
weight
trends
us children
nutrition examination survey
3rd national-health
insulin-resistance
risk-factors
growth
national-health
prevalence
metabolic syndrome
childhood
exposure
united-states
population
blood-pressure
cardiovascular-disease
obesity
nhanes
age
association
hypertension
children
consumption
women
disease
asthma
health
mortality
adults
risk
nutrition
pregnancy
impact
patterns

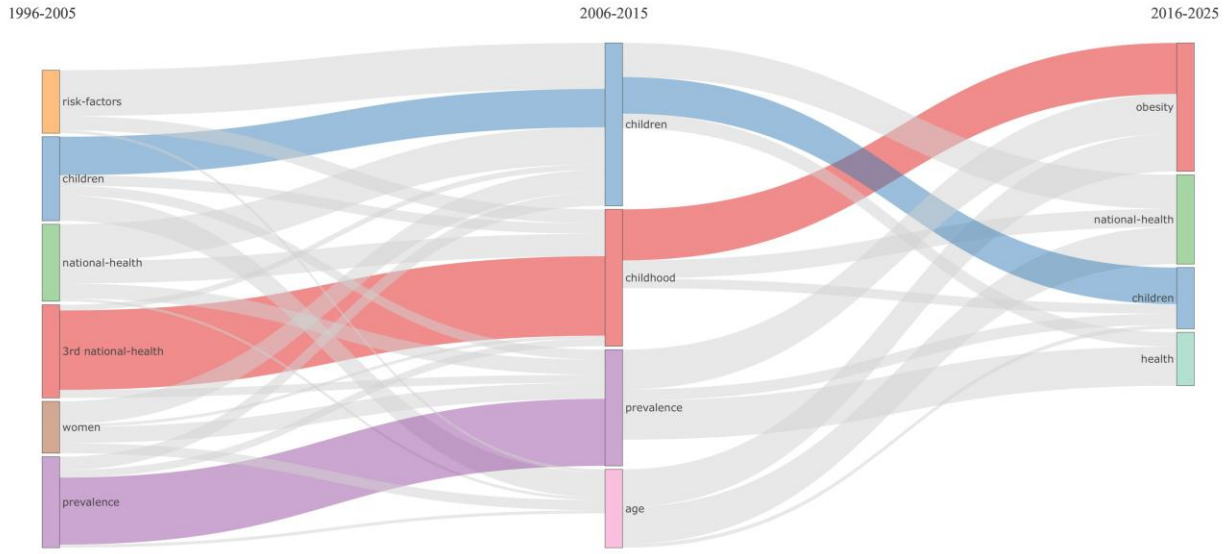

Biomarker utilization ranking
Lead / blood lead
Cholesterol
Insulin
Glucose
Triglycerides
Vitamin D / 25(OH)D
Cotinine
Creatinine
Mercury
Cadmium
CRP / C-reactive protein
PFAS / perfluoro
HbA1c
eGFR
0
100
200
300
400
Number of publications mentioning biomarker
Count
400
300
200
100

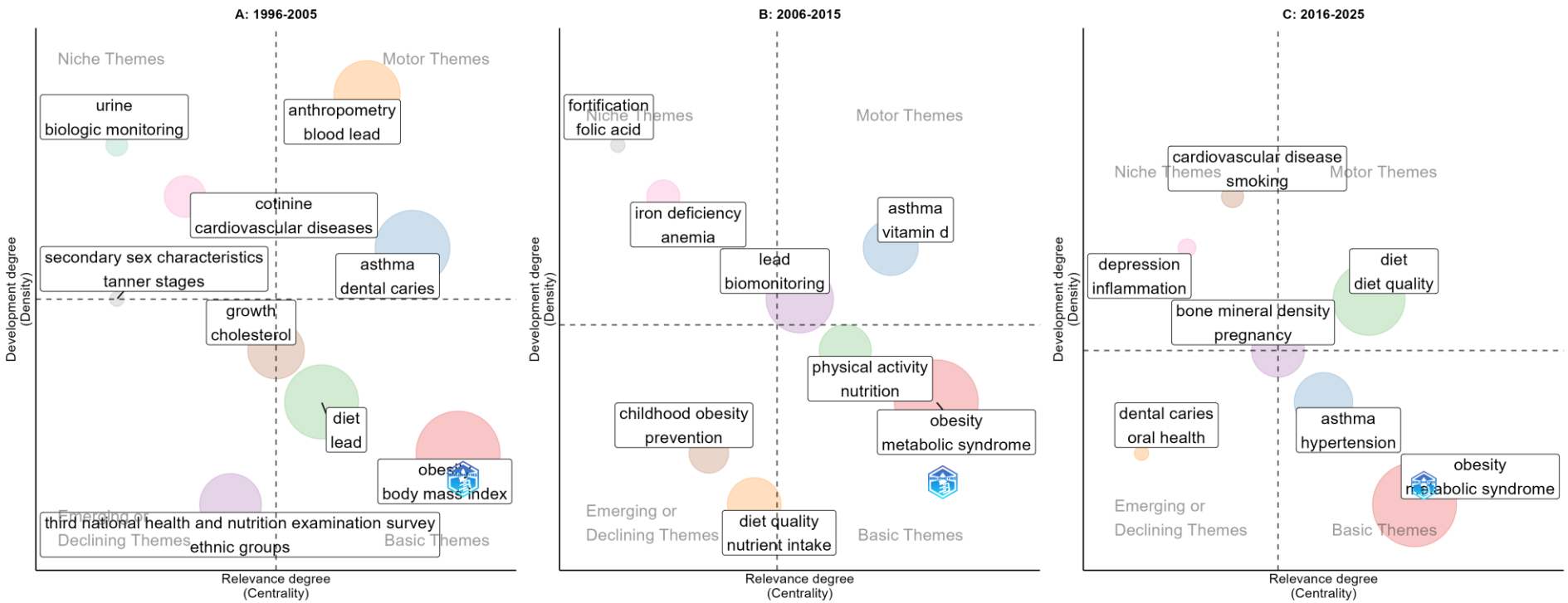

Biomarker evolution across three decades
Total publications per period
0
500
1000
1500
1996-2005
2006-2015
2016-2025
Time period
Cadmium
Cholesterol
Cotinine
Creatinine
CRP
eGFR
Glucose
HbA1c
Insulin
Lead
Mercury
PFAS
Triglycerides
Vitamin D


Biomarker utilization over time
A
B
C
D
Number of publications
Year
0
10
20
30
5
15
2000
2010
2020
A
CRP / C-reactive protein
Vitamin D / 25(OH)D
Lead / blood lead
Mercury
B
Cadmium
PFAS / perfluoro
Cotinine
HbA1c
C
Creatinine
eGFR
Cholesterol
Triglycerides
D
Insulin
Glucose


Biomarker frequency by period
Vitamin D / 25(OH)D
Triglycerides
PFAS / perfluoro
Mercury
Lead / blood lead
Insulin
HbA1c
Glucose
eGFR
CRP / C-reactive protein
Creatinine
Cotinine
Cholesterol
Cadmium
1996-2005
2006-2015
2016-2025
Time slice
Count
250
200
150
100
50
0

Dominant themes per period and quadrant				
Period	**Motor**	**Niche**	**Basic**	**Emerging/Declining**
A: 1996-2005	obesity, asthma, diet, third national health and nutrition examination survey, anthropometry, growth, cotinine	secondary sex characteristics, urine	-	-
B: 2006-2015	asthma	iron deficiency, fortification	obesity, physical activity, lead	diet quality, childhood obesity
C: 2016-2025	-	-	obesity	asthma, diet, bone mineral density, dental caries, cardiovascular disease, depression




A: 1996-2005
Niche Themes
Motor Themes
urine
biologic monitoring
anthropometry
blood lead
cotinine
cardiovascular diseases
secondary sex characteristics
tanner stages
asthma
dental caries
growth
cholesterol
diet
lead
obesity
body mass index
third national health and nutrition examination survey
ethnic groups
Emerging or
Declining Themes
Basic Themes
Development degree
(Density)
Relevance degree
(Centrality)
B: 2006-2015
fortification
folic acid
Niche Themes
Motor Themes
iron deficiency
anemia
asthma
vitamin d
lead
biomonitoring
physical activity
nutrition
childhood obesity
prevention
obesity
metabolic syndrome
Emerging or
Declining Themes
diet quality
nutrient intake
Basic Themes
Development degree
(Density)
Relevance degree
(Centrality)
C: 2016-2025
cardiovascular disease
smoking
Niche Themes
Motor Themes
depression
inflammation
diet
diet quality
bone mineral density
pregnancy
dental caries
oral health
asthma
hypertension
obesity
metabolic syndrome
Emerging or
Declining Themes
Basic Themes
Development degree
(Density)
Relevance degree
(Centrality)